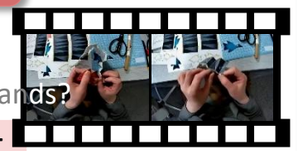
High-level Grounded Reasoning



① Progress Reasoning

Which step has been completed by the hands?

② Hand-Object Grounded CoT

Where is the **tag** after the **hand** pulls it loose from the garment

③ Goal-Oriented Object Tracking

Which body part is interacting with the **dark blue sweater**, and what spatial and temporal evidence confirms this?

Egocentric Grounding & Perception

① Active Object Grounding

At **t1**, which operator part is engaging by holding **a pot**?

② Hand-Object Association

At **t5**, what is the **operator's left hand** interacting with?

③ Manipulation State Perception

At **t2**, what is the manipulation state of the **operator's left hand?**

Example

Question: *Where is the bottle after the sequence ends at 00:06.25, given it was initially located at [201, 1, 402, 316] and manipulated as described?*

A. [0, 410, 300, 816] *B. [648, 225, 818, 424]*
C. [450, 120, 600, 300] *D. [100, 50, 300, 250]*

Generated Reasoning Steps:

Step1: *The start bbox [201, 1, 402, 316] corresponds to the yellow bottle's initial position on the table.*

Step2: *The final bbox [648, 225, 818, 424] at the end time 00:06.25 reflects the yellow bottle's repositioned location after being lifted and placed back down on the scale, so B is correct.*

Data Construction Pipeline

Multi-model Collaborative Perception

Human Verification

Faithful Programmatic Sampling

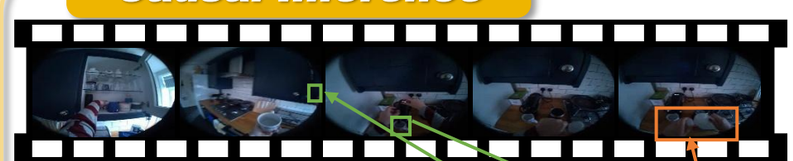
Spatio-Temporal Scene Graph (STSG)

LLM Render

Manual Screening

Question	Metadata
Answer	<timestamp>
Rationale	<bbox>
	<interaction>

Predictive & Causal Inference



① Next State Anticipation

What is the immediate next state of the **cabinet handle**?

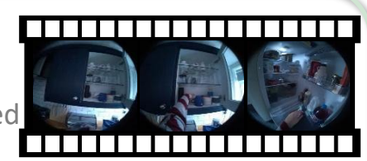
② Next Action Anticipation

What is the operator most likely to do next with their **hand**?

③ Local Cause-Outcome Inference

Which recent action directly caused the **coffee cup** to become partially filled with dark liquid?

Spatio-Temporal Retrospection



① State Retrospection

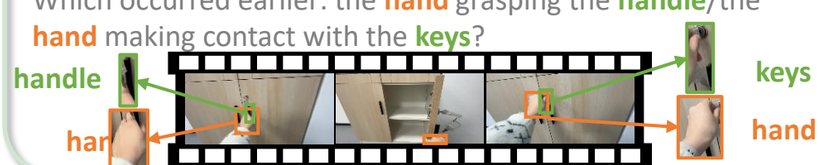
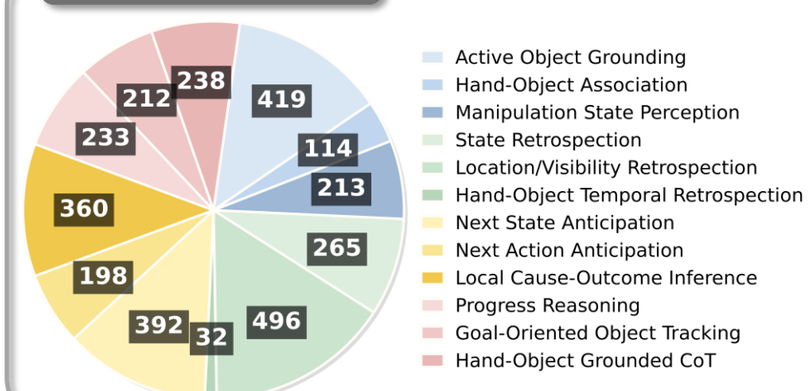
Where was **the cup** before it was grasped by the operator's hand?

② Location / Visibility Retrospection

What was the status of the operator's **hands** immediately before they reached into the refrigerator?

③ Hand-Object Temporal Retrospection

Which occurred earlier: the **hand** grasping the **handle**/the **hand** making contact with the **keys**?



Statistics